

Module 08: SQL Views and Security Abstraction

SQL Views Architecture & Security Abstraction

Virtual Tables, Security Column Masking, Updatability Rules & Materialized Views

1. View Concepts & Architecture

What is a View? (Virtual Table)

- A view stores NO physical data rows on disk!
- Only the SQL query definition is stored in the Data Dictionary.

```
CREATE VIEW IT_Emp AS SELECT EmpID, Name FROM Employee WHERE Dept='IT';
```

3 Fundamental Advantages:

- Security & Data Hiding: Masks confidential columns (Salary, SSN).
- Query Simplification: Hides complex 5-table joins behind a simple view.
- Logical Data Independence: Base table schema changes do not break apps.

```
GRANT SELECT ON IT_Emp TO 'analyst_user';
```

Materialized View (Physical Persistence)

- Physically saves query result on disk (Data Warehouse / OLAP).
- Periodic refresh on schedule: REFRESH COMPLETE ON DEMAND.

Accelerates analytical aggregation queries by 100x!

2. Updatable Views & WITH CHECK OPTION

When is a View Updatable (INSERT/UPDATE)?

A view can be updated IF and ONLY IF:

- It is derived from exactly ONE base table.
- Contains the Primary Key of the underlying table.
- Contains NO aggregate functions (SUM, AVG, COUNT).
- Contains NO GROUP BY, HAVING, or DISTINCT clauses.
- Base table NOT NULL columns without default must be present!

WITH CHECK OPTION (Integrity Guard)

Prevents inserts/updates that violate the view's WHERE predicate!

```
CREATE VIEW Sales_Staff AS SELECT * FROM Emp WHERE Dept = 'Sales' WITH CHECK OPTION;
```

Trying to INSERT an 'HR' employee through Sales_Staff will be REJECTED!

Virtual Abstraction: Views do not store physical rows; they store query definitions providing fine-grained security masking and query simplification.

1. Updatable Views Criteria

A view is updatable only when derived from a single base table without aggregate functions, GROUP BY, DISTINCT, or missing NOT NULL columns.

2. WITH CHECK OPTION

Guarantees that all inserts and updates performed through the view strictly conform to the view's WHERE predicate filter.